

Unit - I

Matrices

Part - A

- ① If 2 & 3 are E. values of $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ find the Eigen values of A^3 & A^{-1}

$$\lambda_1 = 2$$

$$\lambda_2 = 4$$

$$\lambda_3 = 3$$

$$A^3 \rightarrow 2^3 = 8$$

$$4^3 = 64$$

$$3^3 = 27$$

$$A^{-1} \rightarrow \frac{1}{\lambda_1} = \frac{1}{2}$$

$$\frac{1}{\lambda_2} = \frac{1}{4}$$

$$\frac{1}{\lambda_3} = \frac{1}{3}$$

- ② If the E. values of matrix A of order 3×3 are 2, 3, 1 & then find the E. values of adjoint of A.

Wkt $\boxed{\text{adj } A = |A| A^{-1}}$

E. values of A are 2, 3, 1

E. values of A^{-1} are $\frac{1}{2}, \frac{1}{3}, 1$

$|A| = \text{Prod. of E. values}$

$$= 2 \times 3 \times 1$$

$$= 6$$

$$\text{adj } A = 6 \times \frac{1}{2}, 6 \times \frac{1}{3}, 6 \times 1$$

$$= 3, 2, 6$$

- ③ State CH Theorem & uses

→ Every sq. matrix satisfies its own chara equ

Uses

To calculate

i) The integral of A

ii) The inverse of non-singular sq. matrix.

- ④ Quadratic form of the matrix $\begin{bmatrix} 5 & -1 & -2 \\ -1 & 7 & 3 \\ -2 & 3 & 2 \end{bmatrix}$

$$5x_1^2 + 7x_2^2 - 2x_3^2 - 2x_1x_2 - 4x_1x_3 + 6x_2x_3$$

⑤ Expl the nature of Q.F $2x^2 + 3y^2 + 2z^2 + 2xy$

The E. values are 2, 3, 2

Nature is +ve definite.

Part - B

⑦ i) Find the E. values & E. vectors of $\begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$

Sol

$$\text{Let } A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$$

The C.E of 3×3 matrix is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

$S_1 = \text{Sum of main diagonal elements}$

$$= 11 - 2 - 6$$

$$= 11 - 8$$

$$\boxed{S_1 = 3}$$

$S_2 = \text{Sum of minors of diagonal elements}$

$$\begin{vmatrix} -2 & -5 \\ -4 & -6 \end{vmatrix} + \begin{vmatrix} 11 & -7 \\ 10 & -6 \end{vmatrix} + \begin{vmatrix} 11 & -4 \\ 7 & -2 \end{vmatrix}$$

$$(12 - 20) + (-66 + 70) + (-22 + 28)$$

$$-8 + 4 + 6$$

$$\boxed{S_2 = 2}$$

$$S_3 = |A|$$

$$\begin{vmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{vmatrix}$$

$$11(12 - 20) + 4(-42 + 50) - 7(-28 + 20)$$

$$11(-8) + 4(8) - 7(-8)$$

$$= -88 + 32 + 56 \Rightarrow \boxed{S_3 = 0}$$

The CE is $\lambda^3 - 3\lambda^2 + 2\lambda = 0$

$$\lambda[\lambda^2 - 3\lambda + 2] = 0$$

$$\lambda = 0 \mid \lambda^2 - 3\lambda + 2 = 0$$

$$\lambda = 1, 2$$

$$\begin{matrix} & 2 \\ & \swarrow \downarrow \searrow \\ -1 & -3 & -2 \end{matrix}$$

The E.V are 0, 1, 2

To find E. vector

i) $(A - \lambda I)x = 0$

$$\begin{bmatrix} (11-\lambda) & -4 & -7 \\ 7 & (-2-\lambda) & -5 \\ 10 & -4 & (-6-\lambda) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$11x_1 - 4x_2 - 7x_3 = 0 \quad \text{--- (1)}$$

$$7x_1 - 2x_2 - 5x_3 = 0 \quad \text{--- (2)}$$

$$10x_1 - 4x_2 - 6x_3 = 0 \quad \text{--- (3)}$$

from eqn (1) & (2)

$$\begin{matrix} & x_1 & x_2 & x_3 \\ & 11 & -4 & -7 \\ & 7 & -2 & -5 \end{matrix}$$

$$\frac{x_1}{(20-14)} = -\frac{x_2}{(-55+49)} = \frac{x_3}{(-22+28)}$$

$$\frac{x_1}{6} = \frac{x_2}{-6} = \frac{x_3}{6}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \textcircled{x_1}$$

$$\begin{matrix} x_1 & x_2 & x_3 \\ 9 & -4 & -7 \\ 7 & -4 & -5 \end{matrix}$$

Case (ii), $\lambda = 1$

$$\begin{bmatrix} 10 & -4 & -7 \\ 7 & -3 & -5 \\ 10 & -4 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$10x_1 - 4x_2 - 7x_3 = 0 \quad \text{--- (4)}$$

$$7x_1 - 3x_2 - 5x_3 = 0 \quad \text{--- (5)}$$

$$10x_1 - 4x_2 - 7x_3 = 0 \quad \text{--- (6)}$$

from eqn (4) & (5)

$$\begin{matrix} x_1 & x_2 & x_3 \\ 10 & -4 & -7 \\ 7 & -3 & -5 \end{matrix}$$

$$\frac{x_1}{(-20+21)} = -\frac{x_2}{(-50+49)} = \frac{x_3}{(-30+28)}$$

$$\frac{x_1}{-1} = \frac{x_2}{1} = \frac{x_3}{-2}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} \rightarrow \textcircled{x_2}$$

Case (iii) $\lambda = 2$

$$\begin{bmatrix} 9 & -4 & -7 \\ 7 & -4 & -5 \\ 10 & -4 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$9x_1 - 4x_2 - 7x_3 = 0 \quad \text{--- (7)}$$

$$7x_1 - 4x_2 - 5x_3 = 0 \quad \text{--- (8)}$$

$$10x_1 - 4x_2 - 8x_3 = 0 \quad \text{--- (9)}$$

from 7, 8

$$\frac{x_1}{(20-28)} = -\frac{x_2}{(-45+49)} = \frac{x_3}{(-36+28)}$$

$$\frac{x_1}{-8} = \frac{x_2}{4} = \frac{x_3}{-8} \rightarrow \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \rightarrow \textcircled{x_3}$$

ii) Using CH find inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

$$\text{Let } A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

The CE is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

$$S_1 = 2 + 1 + 2$$

$$\boxed{S_1 = 5}$$

$$S_2 = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix}$$

$$= (2-0) + (4-1) + (2-0)$$

$$= 2 + 3 + 2$$

$$\boxed{S_2 = 7}$$

$$S_3 = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix}$$

$$2(2-0) - 1(0) + 1(0-1)$$

$$= 4 - 1$$

$$\boxed{S_3 = 3}$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By CH theorem $A^3 - 5A^2 + 7A - 3I = 0$

Multiply A^{-1} on b.s

$$A^{-1}[A^3 - 5A^2 + 7A - 3I] = 0$$

$$A^2 - 5A + 7I - 3A^{-1} = 0$$

$$A^2 - 5A + 7I = 3A^{-1}$$

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+0+1 & 2+1+1 & 2+1+2 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 2+0+2 & 1+1+2 & 1+0+4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 4 & 5 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 & 5 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 5 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix} = 3A^{-1}$$

$$\begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} = 3A^{-1}$$

$$\frac{1}{3} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} = A^{-1}$$

⑧ i) Find E.V & E.Vectors of $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$S_1 = 2 + 2 + 2$$

$$\boxed{S_1 = 6}$$

$$S_2 = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$

$$= (4-1) + (4-1) + (4-1)$$

$$= 3 + 3 + 3$$

$$\boxed{S_2 = 9}$$

$$S_3 = \begin{vmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 2(4-1) + 1(-2+1) + 1(1-2)$$

$$= 6 - 1 - 1$$

$$\boxed{S_3 = 4}$$

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$\begin{vmatrix} 1 & -6 & 9 & -4 \end{vmatrix}$$

(ii) Using CH theorem, A^{-1} if $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{bmatrix}$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$S_1 = 1 + 2 - 3$$

$$S_1 = 1$$

$$S_2 = \begin{vmatrix} 5 & -4 \\ 7 & -5 \end{vmatrix} + \begin{vmatrix} 1 & -2 \\ 3 & -5 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix}$$

$$= (-25 + 28) + (-5 + 6) + (5 - 4)$$

$$= 3 + 1 + 1$$

$$S_2 = 5$$

$$S_3 = \begin{vmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{vmatrix}$$

$$= 1(-25 + 28) - 2(-10 + 12) - 2(14 - 15)$$

$$= 3 - 4 + 2$$

$$S_3 = 1$$

$$\lambda^3 - \lambda^2 + 5\lambda - 1 = 0$$

By CH Theorem $A^3 - A^2 + 5A - I = 0$

$$A^{-1}[A^3 - A^2 + 5A - I] = 0$$

$$A^2 - A + 5I - A^{-1} = 0$$

$$\begin{bmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & 9 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = A^{-1}$$

$$\begin{bmatrix} 3 & -4 & 2 \\ -2 & 1 & 0 \\ -1 & -1 & 19 \end{bmatrix} = A^{-1}$$

⑨ Reduce Q.F to C.F

$$8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 + 4x_1x_3 - 8x_2x_3$$

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$S_1 = 8 + 7 + 3$$

$$S_1 = 18$$

$$S_2 = \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}$$

$$= (21 - 36) + (24 - 4) + (56 - 36)$$

$$= 5 + 20 + 20$$

$$S_2 = 45$$

$$S_3 = \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix}$$

$$= 8(21 - 16) + 6(-18 + 8) + 2(24 - 14)$$

$$= 40 - 60 + 20$$

$$S_3 = 0$$

$$\lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$\lambda[\lambda^2 - 18\lambda + 45] = 0$$

$$\lambda = 0 \mid \lambda^2 - 18\lambda + 45 = 0$$

$$\lambda = 0, 3, 15$$

$$\begin{array}{c} 45 \\ \swarrow \quad \searrow \\ -3 \quad -15 \end{array}$$

To find E. Vector

case (i)

$$\lambda = 0$$

$$\begin{bmatrix} 8-0 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$8x_1 - 6x_2 + 2x_3 = 0 \quad \text{--- (1)}$$

$$-6x_1 + 7x_2 - 4x_3 = 0 \quad \text{--- (2)}$$

$$2x_1 - 4x_2 + 3x_3 = 0 \quad \text{--- (3)}$$

from (1) & (2)

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ 8 & -6 & 2 \\ -6 & 7 & -4 \end{array}$$

$$\frac{x_1}{(24-14)} = -\frac{x_2}{(-32+12)} = \frac{x_3}{(56-36)}$$

$$\frac{x_1}{10} = \frac{x_2}{20} = \frac{x_3}{20}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \rightarrow (X_1)$$

case (ii) $\lambda = 3$

$$\begin{bmatrix} 8-3 & -6 & 2 \\ -6 & 7-3 & -4 \\ 2 & -4 & 3-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$5x_1 - 6x_2 + 2x_3 = 0 \quad \text{--- (4)}$$

$$-6x_1 + 4x_2 - 4x_3 = 0 \quad \text{--- (5)}$$

$$2x_1 - 4x_2 + 0x_3 = 0 \quad \text{--- (6)}$$

$$\frac{x_1}{(24-8)} = -\frac{x_2}{(-20+12)} = \frac{x_3}{(20-36)}$$

$$\frac{x_1}{16} = \frac{x_2}{8} = \frac{x_3}{-16} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} (X_2)$$

Case (iii) $\lambda = 15$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-7x_1 - 6x_2 + 2x_3 = 0 \quad \text{--- (7)}$$

$$-6x_1 - 8x_2 - 4x_3 = 0 \quad \text{--- (8)}$$

$$2x_1 - 4x_2 - 12x_3 = 0 \quad \text{--- (9)}$$

from (7) & (8)

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ -7 & -6 & 2 \\ -6 & -8 & -4 \end{array}$$

$$\frac{x_1}{(24+16)} = \frac{-x_2}{(28+12)} = \frac{x_3}{(56-36)}$$

$$\frac{x_1}{40} = \frac{-x_2}{40} = \frac{x_3}{20}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \rightarrow \textcircled{X_3}$$

To check orthogonal

$$X_1^T X_2 = 0$$

$$\begin{bmatrix} 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = 2+2-4 = 0$$

$$X_2^T X_3 = 0$$

$$\begin{bmatrix} 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = 4-2-2 = 0$$

$$X_3^T X_1 = 0$$

$$\begin{bmatrix} 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = 2-4+2 = 0$$

E.V $\sqrt{x_1^2 + x_2^2 + x_3^2}$ N.E. vector

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \frac{\sqrt{1+4+4}}{\sqrt{9}} = \frac{3}{3} \quad \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \quad \frac{\sqrt{4+1+4}}{\sqrt{9}} = \frac{3}{3} \quad \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}$$

$$X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \quad \frac{\sqrt{4+4+1}}{\sqrt{9}} = \frac{3}{3} \quad \begin{bmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

$$N = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$N^T = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$D = N^T A N$$

$$\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{matrix} \downarrow \\ \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \end{matrix}$$

$$\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{8}{3} - \frac{12}{3} + \frac{4}{3} & \frac{16}{3} - \frac{6}{3} - \frac{4}{3} & \frac{16}{3} + \frac{12}{3} + \frac{2}{3} \\ -\frac{6}{3} + \frac{14}{3} - \frac{8}{3} & -\frac{12}{3} + \frac{7}{3} + \frac{8}{3} & -\frac{12}{3} - \frac{14}{3} - \frac{4}{3} \\ \frac{2}{3} - \frac{8}{3} + \frac{6}{3} & \frac{4}{3} - \frac{4}{3} - \frac{6}{3} & \frac{4}{3} + \frac{8}{3} + \frac{2}{3} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{matrix} \downarrow \\ \begin{bmatrix} 0 & 2 & 10 \\ 0 & 1 & -10 \\ 0 & -2 & 5 \end{bmatrix} \end{matrix} \rightarrow \begin{bmatrix} 0 & \frac{2}{3} + \frac{2}{3} - \frac{4}{3} & \frac{10}{3} - \frac{20}{3} + \frac{10}{3} \\ 0 & \frac{4}{3} + \frac{1}{3} + \frac{4}{3} & \frac{20}{3} - \frac{10}{3} - \frac{10}{3} \\ 0 & \frac{4}{3} - \frac{2}{3} - \frac{2}{3} & \frac{20}{3} + \frac{20}{3} + \frac{5}{3} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

$$y^T D y = y_1 y_2 y_3 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= 0y_1^2 + 3y_2^2 + 15y_3^2$$

Rank $\rightarrow$ No. of sq. terms = 2

Index $\rightarrow$ No. of +ve sq. terms = 2

Signature $\rightarrow$ No. of +ve term - No. of -ve terms = 2

Nature is +ve semi-definite

10) Reduce q.f to C.F

$$x^2 + y^2 + z^2 - 2xy - 2yz - 2zx$$

$$x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_2x_3 - 2x_3x_1$$

$$A = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

$$CE \text{ is } \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$S_1 = 1 + 1 + 1$$

$$= 3$$

$$S_2 = \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$$

$$= (1-1) + (1-1) + (1-1)$$

$$= 0$$

$$S_3 = \begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix}$$

$$1(1-1) + 1(-1-1) + 1(1+1)$$

$$-2 + 2 = 0$$

$$\lambda^3 - 3\lambda^2 + 0\lambda + 0 = 0$$

State the properties of Jacobian

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{\partial(x, y, z)}{\partial(x, y, z)} = 1$$

Unit - II Multivariable Calculus

① If $u = \frac{x+y}{xy}$ find $\frac{\partial u}{\partial x}$ & $\frac{\partial u}{\partial y}$ ② If $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ then find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$

$$u = \frac{x}{xy} + \frac{y}{xy}$$

$$u = \frac{1}{y} + \frac{1}{x}$$

$$\frac{\partial u}{\partial x} = -\frac{1}{x^2}$$

$$\frac{\partial u}{\partial y} = -\frac{1}{y^2}$$

Sol

$$u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$$

$$\frac{\partial u}{\partial x} = \frac{1}{y} - \frac{z}{x^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{z} - \frac{x}{y^2}$$

$$\frac{\partial u}{\partial z} = -\frac{y}{z^2} + \frac{1}{x}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = x \left(\frac{1}{y} - \frac{z}{x^2} \right) + y \left(\frac{1}{z} - \frac{x}{y^2} \right) + z \left(-\frac{y}{z^2} + \frac{1}{x} \right)$$

$$= \frac{x}{y} - \frac{xz}{x^2} + \frac{y}{z} - \frac{yx}{y^2} - \frac{yz}{z^2} + \frac{z}{x}$$

$$= \frac{x}{y} - \frac{z}{x} + \frac{y}{z} - \frac{x}{y} - \frac{y}{z} + \frac{z}{x} = 0$$

③ State the properties of Jacobians

1. If u & v are the funct of x & y then

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = 1$$

$$\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,s)} = \frac{\partial(u,v)}{\partial(r,s)}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix}$$

④ If $x = r \cos \theta$ then find $\frac{\partial(r,\theta)}{\partial(x,y)}$
 $y = r \sin \theta$

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$= r$$

$$\boxed{\frac{\partial(r,\theta)}{\partial(x,y)} = \frac{1}{r}}$$

⑤ Write necessary & Sufficient condition for max & min of a funct of two variables.

Necessary condition

$$f_x(a,b) = 0$$

$$f_y(a,b) = 0$$

Sufficient condition

$$f_{xx}(a,b) = A$$

$$f_{xy}(a,b) = B$$

$$f_{yy}(a,b) = C$$

Part-B

⑦ i) Find Taylor's series of $x^2y^2 + 2x^2y + 3xy^2$ in Powers of $(x+2)$ & $(y-1)$ upto 3rd.

Sol.

$$h = x - a \quad (-2)$$

$$K = y - b \quad (1)$$

Function	values at $(-2, 1)$
$f = x^2y^2 + 2x^2y + 3xy^2$	$f = (-2)^2(1) + 2(-2)^2(1) + 3(-2)(1)^2$ $= 4 + 8 - 6$ $= 6$
$f_x = 2xy^2 + 4xy + 3y^2$	$f_x = 2(-2)(1)^2 + 4(-2)(1) + 3(1)^2$ $= -4 - 8 + 3$ $= -9$
$f_y = 2yx^2 + 2x^2 + 6xy$	$f_y = 2(1)(-2)^2 + 2(-2)^2 + 6(-2)(1)$ $= 4 + 8 - 12$ $= 0$
$f_{xx} = 2y^2 + 4y$	$f_{xx} = 2 + 4$ $= 6$
$f_{yy} = 2x^2 + 6x$	$f_{yy} = 8 - 12$ $= -4$
$f_{xy} = 4xy + 4x + 6y$	$f_{xy} = 4(-2)(1) + 4(-2) + 6(1)$ $= -8 - 8 + 6$ $= -16 + 6 \Rightarrow -10$
$f_{xxx} = 0$	$f_{xxx} = 0$
$f_{yyy} = 0$	$f_{yyy} = 0$
$f_{xxy} = 4y + 4$	$f_{xxy} = 8$
$f_{xyy} = 4x + 6$	$f_{xyy} = -8 + 6$ $= -2$

Taylor Series

$$f(x, y) = f(a, b) + \frac{1}{1!} [hf_x(a, b) + Kf_y(a, b)]$$

$$+ \frac{1}{2!} [hf_{xx}(a, b) + 2hKf_{xy}(a, b) + K^2f_{yy}(a, b)]$$

$$+ \frac{1}{3!} [h^3f_{xxx}(a, b) + 3h^2Kf_{xxy}(a, b) + 3hK^2f_{xyy}(a, b) + K^3f_{yyy}(a, b)]$$

Where $h = x - a \Rightarrow x + 2$

$K = y - b \Rightarrow y - 1$

② Find the maxima (or) minimum values of $3x^2 - y^2 + x^3$

Sol $f(x, y) = 3x^2 - y^2 + x^3$

$$f_x = 6x + 3x^2$$

$$f_y = -2y$$

$$f_{xx} = 6 + 6x \rightarrow A$$

$$f_{xy} = 0 \rightarrow B$$

$$f_{yy} = -2 \rightarrow C$$

To find stationary points

$$f_x(a, b) = 0$$

$$f_y(a, b) = 0$$

$$6x + 3x^2 = 0$$

$$-2y = 0$$

$$x(6 + 3x) = 0$$

$$\boxed{x=0} \quad 6 + 3x = 0$$

$$\boxed{y=0}$$

$$3x = -6$$

$$\boxed{x=-2}$$

The stationary points are $(0, 0)$, $(-2, 0)$

	$(0, 0)$	$(-2, 0)$
A	6	-6
B	0	0
C	-2	-2
$AC - B^2$	$-12 < 0$	$12 > 0$
Result	Saddle	Max

To find max at $(-2, 0)$

$$\begin{aligned} f(x, y) &= 3x^2 - y^2 + x^3 \\ &= 3(-2)^2 + (-2)^3 \\ &= 12 - 8 \\ &= 4 \end{aligned}$$

⑧ ii) Find the extreme values of the function $x^3 + y^3 - 12x - 3y + 20$

$$f(x, y) = x^3 + y^3 - 12x - 3y + 20$$

To find sta. points

$$f_x = 3x^2 - 12$$

$$f_x(a, b) = 3x^2 - 12 = 0$$

$$f_y = 3y^2 - 3$$

$$3x^2 = 12$$

$$f_{xx} = 6x \rightarrow \textcircled{A}$$

$$x^2 = 4$$

$$f_{xy} = 0 \rightarrow \textcircled{B}$$

$$\boxed{x = \pm 2}$$

$$f_{yy} = 6y \rightarrow \textcircled{C}$$

$$f_y(a, b) = 3y^2 - 3 = 0$$

$$3y^2 = 3$$

$$\boxed{y = \pm 1}$$

$(2, 1)$ $(-2, 1)$ $(2, -1)$ $(-2, -1)$

	$(2, 1)$	$(-2, 1)$	$(2, -1)$	$(-2, -1)$
A	12	-12	12	-12
B	0	0	0	0
C	6	6	-6	-6
$AC - B^2$	$72 > 0$	$-72 < 0$	$-72 < 0$	$72 > 0$
Result	Mini	Saddle	Saddle	Max

To find min at (2,1)

$$f(x,y) = x^3 + y^3 - 12x - 3y + 20$$

$$= 8 + 1 - 24 - 3 + 20$$

$$= 2$$

To find max at (-2,-1)

$$f(x,y) = (-2)^3 + (-1)^3 - 12(-2) - 3(-1) + 20$$

$$= -8 - 1 + 24 + 3 + 20$$

$$= 38$$

Q ii) Exp $e^x \sin y$ in powers of x & y
for as the terms of 3rd degree.

Q i) If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$ find the value of $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} \quad \begin{aligned} \frac{\partial u}{\partial x} &= -\frac{yz}{x^2} & \frac{\partial v}{\partial x} &= \frac{z}{y} \\ \frac{\partial u}{\partial y} &= \frac{z}{x} & \frac{\partial v}{\partial y} &= -\frac{zx}{y^2} \\ \frac{\partial u}{\partial z} &= \frac{y}{x} & \frac{\partial v}{\partial z} &= \frac{x}{y} \end{aligned}$$

$$\frac{\partial w}{\partial x} = \frac{y}{z}$$

$$\frac{\partial w}{\partial y} = \frac{x}{z}$$

$$\frac{\partial w}{\partial z} = -\frac{xy}{z^2}$$

$$= \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix}$$

$$= -\frac{yz}{z^2} \left[-\frac{zx}{y^2} \left(-\frac{xy}{z^2} \right) - \frac{x}{z} \left(\frac{x}{y} \right) \right] - \frac{z}{x} \left[\frac{z}{y} \left(-\frac{xy}{z^2} \right) - \frac{x}{y} \left(\frac{y}{z} \right) \right] + \frac{y}{z} \left[\frac{z}{y} \left(\frac{x}{z} \right) + \frac{zx}{y^2} \left(\frac{y}{z} \right) \right]$$

$$= -\frac{yz}{z^2} \left[\frac{x^2}{y^2} - \frac{x^2}{yz} \right] - \frac{z}{x} \left[-\frac{x}{z} - \frac{x}{z} \right] + \frac{y}{x} \left[\frac{x}{y} + \frac{x}{y} \right]$$

$$= -\frac{z}{x} \left[-\frac{2x}{z} \right] + \frac{y}{x} \left[\frac{2x}{y} \right]$$

$$= 2 + 2$$

$$= 4$$

Q ii) A rectangular box open at the top, is to have a volu of 108cc. Find the dimensions of the box that requires the least material for its construction.

10) i) If $x+y+z=u$ then prove that $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$
 $y+z=uv$
 $z=uvw$

sol

$$\begin{array}{l|l|l} x+uv=u & y+uvw=uv & z=uvw \\ x=u-uv & y=uv-uvw & \\ \boxed{x=u(1-v)} & \boxed{y=uv(1-w)} & \end{array}$$

$$\begin{array}{l} x=u(1-v) \\ =u-uv \\ y=uv(1-w) \\ =uv-uvw \\ z=uvw \end{array} \quad \begin{array}{l} \frac{\partial x}{\partial u} = 1-v \\ \frac{\partial x}{\partial v} = -u \\ \frac{\partial x}{\partial w} = 0 \end{array} \quad \begin{array}{l} \frac{\partial y}{\partial u} = v-vw \\ =v(1-w) \\ \frac{\partial y}{\partial v} = u-uw \\ =u(1-w) \\ \frac{\partial y}{\partial w} = -uv \end{array} \quad \begin{array}{l} \frac{\partial z}{\partial u} = vw \\ \frac{\partial z}{\partial v} = uw \\ \frac{\partial z}{\partial w} = uv \end{array}$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 1-v & -u & 0 \\ v(1-w) & u(1-w) & -uv \\ vw & uw & uv \end{vmatrix}$$

$$= (1-v) \left[(u(1-w))uv + uv(uw) \right] + u \left[(v(1-w))uv + uv(vw) \right] + 0$$

$$= (1-v) \left[(u-uw)(uv) + u^2vw \right] + u \left[(v-vw)uv + uv^2w \right]$$

$$= (1-v) \left[u^2v - u^2vw + u^2vw \right] + u \left[uv^2 - uv^2w + uv^2w \right]$$

$$= (1-v) \left[u^2v \right] + u \left[uv^2 \right]$$

$$= u^2v - u^2v + u^2v$$

$$= u^2v$$

Hence Proved.

① The temp $u(x, y, z)$ at any point in space is $u = 400xyz^2$. Find the highest temp on surface of the sphere $x^2 + y^2 + z^2 = 1$

Unit - III Vector Calculus

Part - A

① Find a unit normal vector to the surface $x^2 + y^2 = z$ at $(1, -2, 5)$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k} \quad x^2 + y^2 - z = 0$$

$$\nabla\phi(1, -2, 5) = 2(1)\vec{i} + 2(-2)\vec{j} - \vec{k}$$

$$= 2\vec{i} - 4\vec{j} - \vec{k}$$

$$|\vec{n}| = \sqrt{4 + 16 + 1}$$

$$= \sqrt{21}$$

$$\vec{u} = \frac{2\vec{i} - 4\vec{j} - \vec{k}}{\sqrt{21}}$$

② Find the D.D of $\phi = x^2 + y^2 + z^2$ at $(1, 1, 1)$ in the direction of $\vec{i} + 2\vec{j} + 2\vec{k}$

$$\boxed{D.D = \nabla\phi \cdot \frac{\vec{a}}{|\vec{a}|}}$$

$$\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

$$= 2x\vec{i} + 2y\vec{j} + 2z\vec{k}$$

$$\nabla\phi(1, 1, 1) = 2\vec{i} + 2\vec{j} + 2\vec{k}$$

$$|\vec{a}| = \sqrt{1^2 + 2^2 + 2^2}$$

$$= \sqrt{1 + 4 + 4}$$

$$= \sqrt{9}$$

$$= 3$$

$$D.D = \frac{2\vec{i} + 2\vec{j} + 2\vec{k} \cdot (\vec{i} + 2\vec{j} + 2\vec{k})}{3}$$

$$= \frac{2 + 4 + 4}{3}$$

$$= \frac{10}{3}$$

- ③ Find the value a so that the vector $\vec{F} = (x+3y)\vec{i} + (y-2z)\vec{j} + (x+az)\vec{k}$ is solenoidal.

$$\nabla \cdot \vec{F} = 0$$

$$\Rightarrow \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+az)$$

$$1 + 1 + a = 0$$

$$\boxed{a = -2}$$

- ④ If $\vec{F} = x^2\vec{i} + y^2\vec{j}$ evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the line $y=x$ from $(0,0)$ to $(1,1)$

- ⑤ State Green's theorem. If $M, N, \frac{\partial M}{\partial y}, \frac{\partial N}{\partial x}$ are continuous and single valued function in the region R enclosed by the curve C , then
- $$\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

Past - B

- ⑦ i) Find the angle b/w the surfaces $x^2 - y^2 - z^2 = 11$ & $xy + yz - zx = 18$ at the point $(6, 4, 3)$

$$\cos \theta = \frac{|\nabla \phi_1 \cdot \nabla \phi_2|}{|\nabla \phi_1| |\nabla \phi_2|}$$

$$x^2 - y^2 - z^2 - 11 = 0$$

$$xy + yz - zx - 18 = 0$$

$$\phi_1 = x^2 - y^2 - z^2 - 11$$

$$\nabla \phi_1 = \frac{\partial \phi_1}{\partial x} \vec{i} + \frac{\partial \phi_1}{\partial y} \vec{j} + \frac{\partial \phi_1}{\partial z} \vec{k}$$

$$= 2x\vec{i} - 2y\vec{j} - 2z\vec{k}$$

$$\nabla \phi_1(6, 4, 3) = 12\vec{i} - 8\vec{j} - 6\vec{k}$$

$$|\nabla \phi_1| = \sqrt{12^2 + 8^2 + 6^2}$$

$$= \sqrt{144 + 64 + 36}$$

$$= \sqrt{244}$$

$$\phi_2 = xy + yz - zx - 18$$

$$\nabla \phi_2 = (y-z)\vec{i} + (x+z)\vec{j} + (y-x)\vec{k}$$

$$\nabla \phi_2(6,4,3) = (4-3)\vec{i} + (6+3)\vec{j} + (4-6)\vec{k}$$

$$= \vec{i} + 9\vec{j} - 2\vec{k}$$

$$|\nabla \phi_2| = \sqrt{1+81+4}$$

$$= \sqrt{86}$$

$$\text{Sol } \nabla \phi_2 = 12\vec{i} - 8\vec{j} - 6\vec{k} \cdot \vec{i} + 9\vec{j} - 2\vec{k}$$

$$\sqrt{244} \sqrt{86}$$

$$= 12 - 72 + 12 = -48$$

$$\sqrt{244} \sqrt{86}$$

ii) show that $r^n \vec{r}$ is irrotational vector for all values of n but is solenoidal only if $n = -3$

$$\text{Sol } \vec{F} = r^n \vec{r}$$

$$= r^n [x\vec{i} + y\vec{j} + z\vec{k}]$$

$$\vec{F} = x r^n \vec{i} + y r^n \vec{j} + z r^n \vec{k}$$

To prove $\vec{F}$ is irrotational

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x r^n & y r^n & z r^n \end{vmatrix}$$

$$= \sum \vec{i} [\frac{\partial}{\partial y} z r^n - \frac{\partial}{\partial z} y r^n]$$

$$= \sum \vec{i} [z \cdot n r^{n-1} \frac{\partial r}{\partial y} - y \cdot n r^{n-1} \frac{\partial r}{\partial z}]$$

$$= \sum \vec{i} [z \cdot n r^{n-1} (\frac{y}{r}) - y \cdot n r^{n-1} (\frac{z}{r})]$$

$$= \sum \vec{i} [z n r^{n-2} y - y n r^{n-2} z]$$

$$= \vec{0}$$

To prove $\vec{F}$ is solenoidal for $n = -3$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x} (x r^n) + \frac{\partial}{\partial y} (y r^n) + \frac{\partial}{\partial z} (z r^n)$$

$$= \sum [x \cdot n r^{n-1} \frac{\partial r}{\partial x} + r^n (1)]$$

$$= \sum [x \cdot n r^{n-1} (\frac{x}{r}) + r^n]$$

$$= \sum [x^2 \cdot n r^{n-2} + r^n]$$

$$= \sum x^2 n r^{n-2} + \sum r^n$$

$$n r^{n-2} [x^2 + y^2 + z^2] + 3 r^n$$

$$n r^{n-2} \cdot r^2 + 3 r^n$$

$$n r^n + 3 r^n$$

$$= r^n (n+3)$$

$\vec{F}$ is solenoidal only if $n = -3$

⑧ verify Gauss divergence theorem

for $\vec{F} = 4xz\vec{i} + y^2\vec{j} + z^2\vec{k}$ take over the cube bounded by the planes.

$$x=0, x=1, y=0, y=1 \text{ and } z=0, z=1$$

⑨ Verify Gauss divergence theorem for

$$\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k} \text{ taken over}$$

the cube bounded by $x=0, x=a, y=0, y=b, z=0$ and $z=c$.

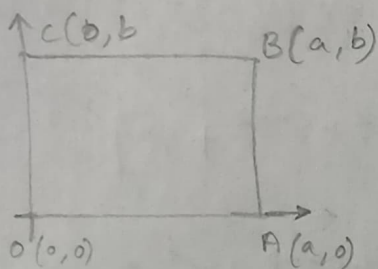
$$\iint_S \vec{F} \cdot \vec{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv$$

⑩ i) Verify Stokes's theorem for

$\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ where S is the rectangle in the xy -plane formed by the lines $x=0, x=a, y=0, y=b$.

Stokes's theorem

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$



Surface	$\nabla \times \vec{F}$ Curl $\vec{F}$	$\vec{n}$	$\text{curl } \vec{F} \cdot \vec{n}$	ds
OABC	$4y\vec{k}$	$\vec{k}$	$4y$	$dx dy$

✓ Show that $\vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational and find its scalar potential.

To prove irrotational

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (6xy + z^3) & (3x^2 - z) & (3xz^2 - y) \end{vmatrix}$$

Unit - IV

Part - A

① Verify $f(z) = z^3$ is analytic or not.

② State Cauchy's theorem $\int_C f(z) dz = 0$

③ What is the value of the integral

$$\int_C \frac{5z^2 + 14z + 1}{z + 2} dz \text{ where } |z| = \frac{1}{2}$$

④ Expand $f(z) = \cos z$ in Taylor series abt the origin.

⑤ Identify & classify the singularities of the funct $\frac{1}{e^z + 1}$

Unit - V

Ordinary differential Equations

① Solve $(D^3 - 5D^2 + 8D - 4)y = 0$
The AE is $m^3 - 5m^2 + 8m - 4 = 0$

$$1 \mid \begin{array}{rrrr} -5 & 8 & -4 \\ & 1 & -4 & 4 \\ & 1 & -4 & 4 \\ \hline & & & 0 \end{array}$$

$$m_1 = 1 \quad m^2 - 4m + 4 = 0$$

$$m_2 = m_3 = 2 = m$$

$$\begin{array}{c} +4 \\ \swarrow \quad \searrow \\ -2 \quad -2 \end{array}$$

$$CF = Ae^{m_1 x} + (Bx + C)e^{m_2 x} = Ae^x + (Bx + C)e^{2x}$$

$$PI = \frac{RHS}{f(D)} = 0$$

The general soln is $y = CF + PI$